

Entry-Path & Gap Register

PREPARED BY Energy Hippo implementation team

The rule that governs this whole project: the synthetic customer is created by **entering data through the EEM UI** (by hand, or by a system that drives the UI), **not** by inserting into the database. Data may only arrive off-UI through **legitimate product paths** — file / ODBC / handheld gateways, the bill importer, external integrations — or as **side effects of using the app** (audit, status, logs). A blueprint is only useful if it can be produced this way. Where the UI **can't** create something the product otherwise supports, that is a **finding**: either we missed a screen, or the product needs a feature added.

This register re-frames every phase around **how each thing actually gets in**, and lists the gaps. It is grounded in a sweep of the EEM apps + web frontend (WPF DBAdmin, ConfigurationTool, BillImporter; the Angular/Kendo web SPA; the Tasks engine).

Instance assumption

We target a **fully-provisioned EEM instance** (all product stored procs, lookup catalogs, and report-menu registrations present — i.e. a real install, not a from-git build). Several capabilities the sweep flagged as "ships only in the .bak, not in git" (the `BillAllocationMethod`/`PtAllocationMethod` catalogs, the `FF_wn*` regression procs, the reconciliation report registration) are therefore **present** for us. Installing the product's own SQL objects is **standing up EEMSuite**, not seeding customer data — it does not violate the rule. The true gaps below are where **no UI/data-path exists for the customer data even in a full instance**.

Entry-method legend

- **UI-DIRECT** — a screen where you type it in.
- **UI-THEN-RUN** — configure in the UI, then run a task/job/process that produces the result (the result is computed, not typed).
- **DATA-PATH** — enters via a real ingestion path (file/ODBC/handheld/bill import).
- **EXTERNAL** — comes from an outside service/API (ENERGY STAR PM, EnergyAI).
- **SYSTEM-GENERATED** — appears as a side effect of using the app (audit, status, notes, logs).
- **GAP** — the product supports the end state but there is no UI/data-path to create it: **missed-screen** (backend exists, view missing) or **needs-feature** (no path at all).

What IS hand-enterable (the good news)

Most of the plan is straightforwardly UI-creatable in a full instance:

Phase	UI-enterable via
0 Foundation	Companies/Users/Groups (WPF), Company Calendar (web #/CompanyCalendar), permissions; node hierarchy (WPF Nodes)
1 Cost core	Bill Entry (#/BillEntry), Bill Validation rules (WPF BillValidationView), notes/status (system), digital points via point definitions; interval/bill data via DATA-PATH (gateways/import)
2 Rates	WPF Rate Modeler creates providers, rates, classes, determinants of all 9 types (100–800; 210/600 confirmed present in the UtilityDeterminantType lookup), blocks/tiers, TOU/season/holiday schedules, ratchets, load factor, pre/post scripts, effective dates, commodity maps, multi-currency; Rate Import file for bulk. RTP determinant references a price point (point created in Nodes; price values via DATA-PATH).
3 AP/GL	Bill entry/import (UI+DATA-PATH), GL accounts (WPF GLAccountsView), account AP/expense account assignment (WPF), AP approval + run an existing export format (#/reporting/APApprovalAndUpload)
4 Tenant	Bill allocation config (web #/reporting/BillAllocationConfiguration), WUI master-point config (web), bill generation config (WPF TenantRebillView), meter-allocation run (#/reporting/ProcessMeterAllocations)
5 Sustainability	Emission sources/factors(+history values)/meter-assignment/company-config (web #/EmissionFactorSources , #/EmissionFactors , #/MeterEmissionFactorAssignment , #/GHGCompanyConfiguration); ENERGY STAR property setup + rating request (web, local , no PM account); weather regression run (web Custom views)
6 Operations	Project Tracking (web editor), alarm thresholds (WPF PointAlarmDefinitionView), Task scheduler (web #/tasks/manager), Workflow author (WPF ConfigurationTool), MFR save/dashboard/delivery incl. groups for 100+ recipients , Handheld config + export, MyUpload/MyImport; audit/status/notes (system-generated)

The gaps (where the UI can't do it)

Grouped by disposition. **Impact** = which reports/goals are affected.

A. Needs a product feature (no UI path even in a full instance)

#	Gap	Why	Impact	Recommended disposition
A1	AP/GL export FORMAT definition	No create/edit-format UI anywhere; formats are devauthored procs (rp_FinancialInterfaceUpload-style) + Exports_Get rows (the CustomDevelopment\<customer> model). Only running an existing format is UI.	Phase 3 "7 formats." Only the 2 built-ins (export_ap, export_ap2) are usable by hand.	Use the 2 built-ins for the synthetic customer; treat new formats as dev work (or a candidate product feature: a format-builder UI). Flag, don't fake.
A2	Accruals (create + reverse)	No UI/API/service writes Accrual_Bill/BillAccrualMap; the rp_OpenAccruals @Reverse proc isn't wired to any screen.	Phase 3 accruals; Phase 4 tenant accruals; OpenAccruals/accruals/mpaacruals reports.	Needs-feature. Cannot be handentered. Either commission an accrual-entry UI, or accept these reports stay empty. Decision needed.
A3	GL user-defined fields (5 segments) + values	Domain model + API + WPF MVVM models exist, but no XAML view is wired.	Phase 3 GL UDV coding; rp_GLAccountUDV; GL-coded export segments.	Missed-screen — closest to "finish the view." Small dev. Until then, GL coding is base account only.
A4	Per-meter rate-class / rate-class-OPTION binding	The meter ↔ rate assignment (TenantRebillView → UtilityMeterRates) has only RateId + dates; no RateClassId/OptionId field.	Phase 2 #9 (voltage/phase/tax-exempt/farm per meter); rp_RateClassAssignment depth.	Needs-feature. Options can be defined on a rate but not bound to a meter via UI (engine applies defaults). Model the rate with options; note the binding gap.
A5	Non-Unique rate classes (e.g. the always-applied "All" class)	RateClassType hardcoded to Unique on create.	Phase 2 tax/min-bill scoping that relies on an "All" class.	Needs-feature (these are systemseeded classes). Use memberscoping where possible; flag.
A6	Benchmark definitions / EUI targets	Benchmark UI is read-only end-to-end; no create screen.	Phase 5 benchmarks; rp_BenchmarkRanking custom defs.	Needs-feature or rely on built-in benchmark master data. Decision needed.
A7	MFR delivery external-email recipients	Domain Recipient supports email, but the web picker exposes only users + groups.	Phase 6 external recipients.	Missed-screen. Reach 100+ via groups (fully supported); external email needs the field added.

#	Gap	Why	Impact	Recommended disposition
A8	GreenButton (ESPI XML) import	Read-only API only; legacy upload tasks skipped in 7.0; no import UI.	Phase 6 GreenButton intake.	Out-of-scope / needs-feature. Drop GreenButton from the synthetic scope, or flag for reenale.

B. External dependency (legitimate, but needs an outside service)

#	Item	Why	Impact	Disposition
B1	ENERGY STAR scores (ESResponseFacility)	Property setup + rating request are local UI, but scores only arrive from the Portfolio Manager API via the EnergyStar sync task. No hand-entry.	Phase 5 RatingHistory/Comparison/RequestLog score columns; PerformanceSnapshot ES score.	Decision needed: (i) connect a real/sandbox PM account + run the sync task (the legitimate path), (ii) accept ES score reports stay empty (property/request data still demos), or (iii) request a manual-score-entry feature.
B2	EnergyAI insights	Only from the external EnergyAI cloud service (enroll → run → download). Even enrollment is DB-only (no UI).	Phase 6 rp_EnergyAIInsights.	Decision needed: connect the EnergyAI service, or accept this report stays empty. Lowest priority.

C. UI-THEN-RUN — produced by running the product (legit; requires the engine/jobs running)

These are **not** gaps — the data is computed by running EEM, which is exactly how a real customer gets it. They require the **Tasks EngineWorker / SQL Agent jobs** to be running in the target instance.

Item	Configure in UI	Produced by
Tenant bill generation	gen type/schedule (WPF TenantRebillView) + allocation config (web)	SQL Agent job Job_BillAllocation (no on-demand "Generate Bills" button — meterallocation has one; full bill-gen is scheduled)
Alarm events	thresholds (WPF PointAlarmDefinitionView) + load point data	the alarm evaluation/notification task raises rows
Weather regression results	period/config (web Custom views)	running the FF_wn* regression proc
AP interface files	approve bills, pick existing format	running AP Upload (writes ExportFile)
Task logs / gateway status	create schedules / run-now (web #/tasks/manager)	the EngineWorker executing tasks → WorkflowRun/TaskQuickRun

D. DATA-PATH — via real ingestion (legit, the intended path)

Interval & meter data (gateways: AcquiSuite/MV90/BACnet/Modbus/ODBC/NOAA/Aeris/ Spinwave/FIG/Neptune/MVRS/Fake), monthly bills (Bill Importer BIF/CSV/Excel), handheld routes (Neptune/MVRS upload → HMR), and the **RTP hourly price point's values**. These are already the backbone of the existing dataset.

E. SYSTEM-GENERATED — free by using the app

Audit, activity, status history, notes, workflow audit — populated by performing the actions in the UI (bill entry/approval, edits, report runs, logins, including the disabled-user failed-auth and cross-company auditor cases). On by default.

Decisions (resolved 2026-06-26) — close the gaps

The owner chose to **close** the gaps rather than defer them, so the dream customer exercises everything:

1. **ENERGY STAR scores (B1): ✓ connect a sandbox/real PM account** + run the sync task (the only legitimate way to populate scores).
2. **AP export formats (A1): ✓ commission dev** for the richer ERP builder procs (PeopleSoft AP voucher + GL journal, SAP FI, GL-UDF, AR).
3. **All other gaps (A2-A8, B2): ✓ build the features** (accruals, per-meter rate options, non-Unique class, benchmarks, GL-UDF view, MFR external email, EnergyAI, GreenButton).

These become a **product backlog for the EEMSuite repo** — specified with build locations and effort in [product-gap-backlog.pdf](#). The classification still holds (A3/A7 missed-screens; A1/A2/A4/A5/A6/A8 needs-feature; B1/B2 external) — it now drives **what to build/connect**, not **whether**. Everything not in the gap list is hand-enterable (UI-DIRECT) or legitimately produced (RUN / DATA-PATH / SYSTEM-GENERATED) and proceeds to materialize.